Assignment No	**10**
Name	**Satish Rathod**
Title	**Circular Linked List**
Objective	**Insertion and Deletion**
Roll No	**MCA2649**

Source Code:-

```
//Name:- Satish Rathod
//Roll No:- 2649
//LinkedList


// 2) Circular Linked List
// Design a Node class
class Node
{
        int data;
        Node next;
        Node(int data)
        {
                this.data=data;
                this.next=null;
        }
}
```

```
class CircularLinkedList
{
        //add a node
        private Node last; // last.next is head
        public void insertFirst(int data)
        {
                Node n = new Node(data);
                if(last == null) // list is Empty
                {
                        last = n;
                        last.next = last;
                }
                else
                {
                        n.next = last.next;
                        last.next = n;
                }

        }

        // display method
        public void display()
        {
                if(last == null)
                {
```

```
                System.out.println("List is Empty!!");
        }
        Node temp=last.next;
        do
        {
                System.out.print(temp.data+"->");
                temp = temp.next;
        }while(temp != last.next);
        System.out.println("head");
}


// insert last
public void insertLast(int data)
{
        insertFirst(data);
        last = last.next;
}

//count of elements
public int count()
{
        int count = 0;
        if(last==null)
        {
```

```
                return count;
        }
        Node temp = last.next;
        do
        {
                count++;
                temp = temp.next;
        }while(temp != last.next);
        return count;

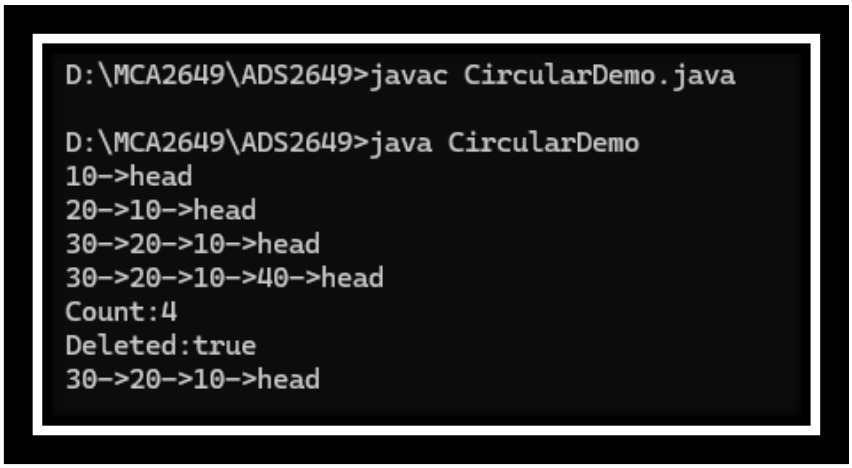
}

//delete
public boolean delete(int key)
{
        if(last==null)
        return false;
        Node curr = last.next;
        Node prev = last;
        do
        {
                if(curr.data == key)
                {
                        prev.next = curr.next;
                        return true;
```

```
                }
                prev=curr;
                curr = curr.next;
            }while(curr != last.next);
            return false;
        }

}
class CircularDemo
{
        public static void main(String args[])
        {
            CircularLinkedList l1 = new CircularLinkedList();
                                                l1.insertFirst(10);
            l1.display();
            l1.insertFirst(20);
            l1.display();
            l1.insertFirst(30);
            l1.display();
            l1.insertLast(40);
            l1.display();
            System.out.println("Count:"+l1.count());
            System.out.println("Deleted:"+l1.delete(40));
            l1.display();
        }
}
```

OUTPUT:-


D:\MCA2649\ADS2649>javac CircularDemo.java
D:\MCA2649\ADS2649>java CircularDemo
10->head
20->10->head
30->20->10->head
30->20->10->40->head
Count:4
Deleted:true
30->20->10->head